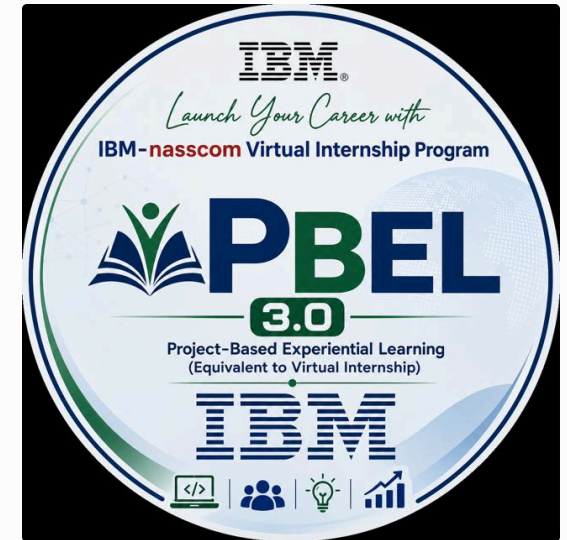


Face Recognition Based Attendance System

Leveraging Artificial Intelligence and Computer Vision for Smart Attendance Management



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Powered By: IBM SkillsBuild • Edunet Foundation • AICTE

Face Recognition Based Attendance System

Python GUI Integrated Smart Attendance System Using Face Recognition





Why Attendance Systems Need a Revolution

10–15 Min Lost

Roll-call wastes precious lecture time every single session

Proxy Fraud

Students mark attendance for absent peers, undermining integrity

Paper Errors

Mismarks, lost registers, no digital backup or real-time access

Smart Alternative

Face recognition offers contactless, intelligent, automated tracking



The Problem with Manual Attendance

Time-Consuming

Roll-call disrupts valuable lecture time every day

Proxy Fraud

Students mark attendance for absent peers with ease

Human Error

Mismarks and lost registers are alarmingly common

Not Scalable

Completely unmanageable in classrooms of 100+ students

Project Objectives



Automate Marking

Zero manual input — attendance captured instantly by camera



Prevent Proxy

Biometric face verification eliminates impersonation



Improve Accuracy

LBPH algorithm delivers reliable recognition results



Save Time

Full class attendance marked in under 30 seconds



Paperless Records

Automatic CSV report generation with timestamps



User-Friendly GUI

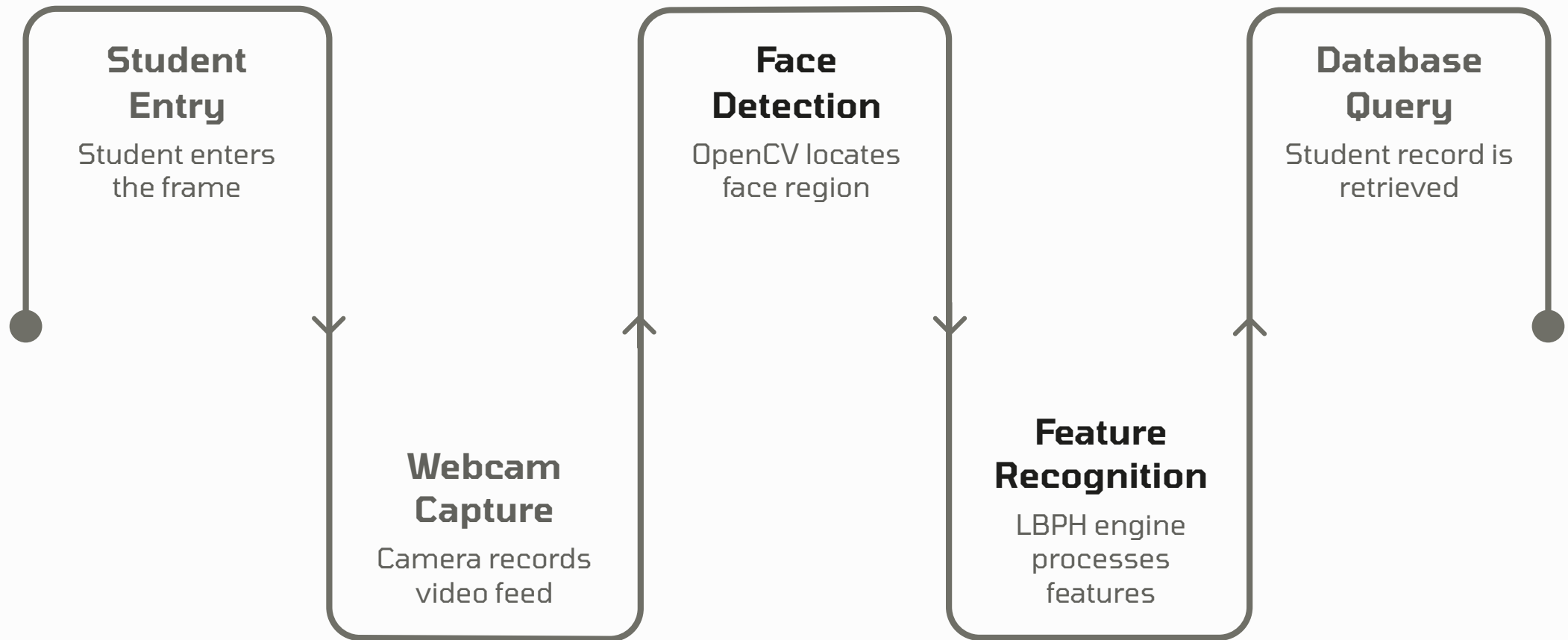
Python Tkinter dashboard for all operations

Technologies Powering the System



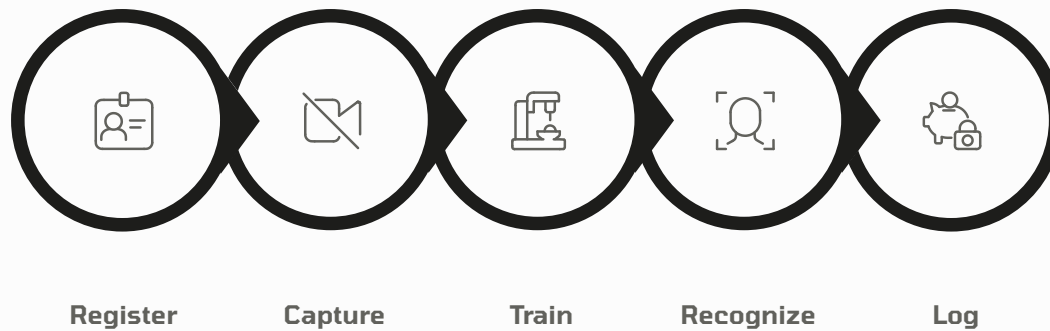
Technology	Role
Python	Core Programming Language
Tkinter	GUI Development
OpenCV	Face Detection & Recognition
LBPH Recognizer	Face Recognition Algorithm
NumPy	Numerical Operations
Pandas	Data Management
CSV	Data Storage
Datetime	Date & Time Logging

System Architecture



End-to-end pipeline from student entry to automated attendance record — zero manual steps required.

Project Workflow



How It Works — Step by Step

01

Register & Capture

Student ID, name, and webcam training images collected

02

Train LBPH Model

One-click training builds the recognition dataset

03

Live Recognition

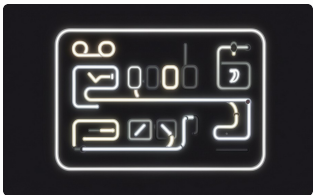
Camera detects, matches face in real-time

04

Log & Display

Attendance saved to CSV and shown on dashboard instantly

GUI Overview



Main Dashboard

Central control panel with navigation to all modules



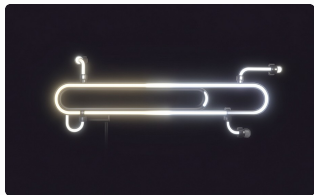
Student Registration

Form with ID, name, and department entry fields



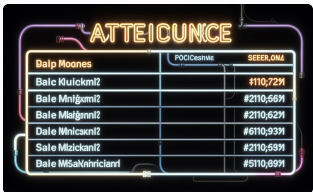
Capture Images

Live webcam feed with real-time face bounding box



Train Model

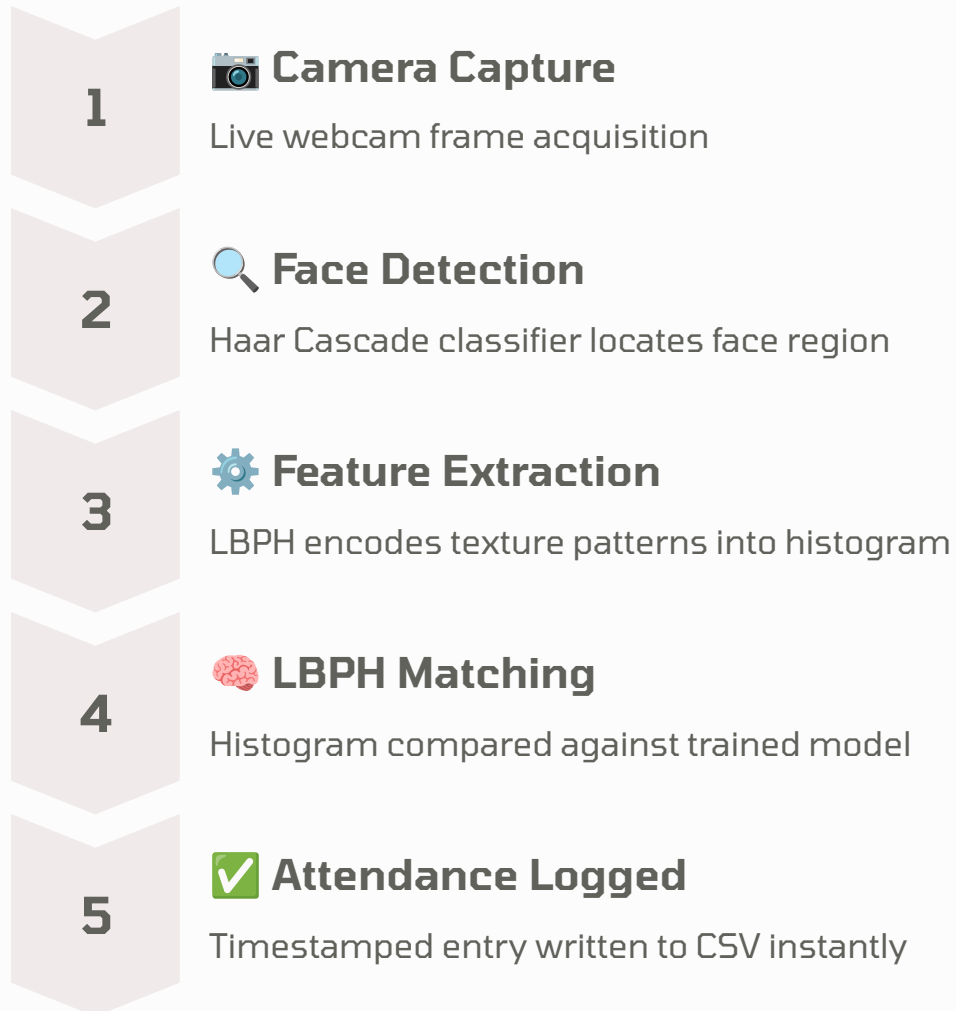
One-click LBPH training with live progress indicator



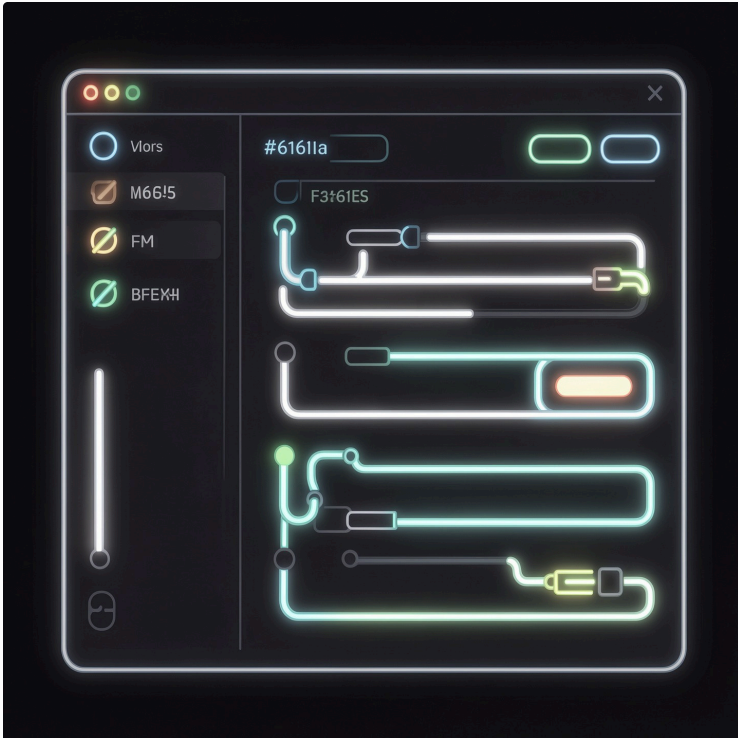
Attendance Window

Live table with name, ID, date, and time columns

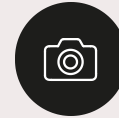
Face Recognition Process



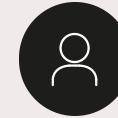
Key Features



**Password-
Protected
Registration**



**Automatic
Real-Time Face
Detection**



**Live
Recognition
with
Confidence
Score**



**Instant
Attendance
Table Updates**



**Automatic CSV
Report
Generation**



**One-Click LBPH
Model Training**

Advantages Over Traditional Systems



Fast

Marks attendance in under **2 seconds** per student



Accurate

LBPH achieves **95%+** recognition accuracy



Contactless

Zero physical interaction required at any stage



Secure

Biometric identity is impossible to forge



Paperless

Fully digital, auto-generated CSV-based records



Proxy-Proof

Physical face presence is absolutely mandatory



Time-Saving

Frees **10+ minutes** of class time every day



Low Maintenance

Minimal hardware requirements, easy software updates



This system transforms classroom attendance from a manual burden into an intelligent, automated process — saving time, eliminating fraud, and empowering educators.

Transforming Attendance Management

The Face Recognition Based Attendance System delivers a modern, efficient, and secure solution that directly addresses the challenges of traditional methods, bringing significant value to educational institutions.



Time Efficiency

Recapture valuable lecture time by automating attendance in seconds, freeing up minutes each session for teaching and learning.



Enhanced Integrity

Eliminate proxy attendance and ensure highly accurate, fraud-proof records through biometric facial verification.



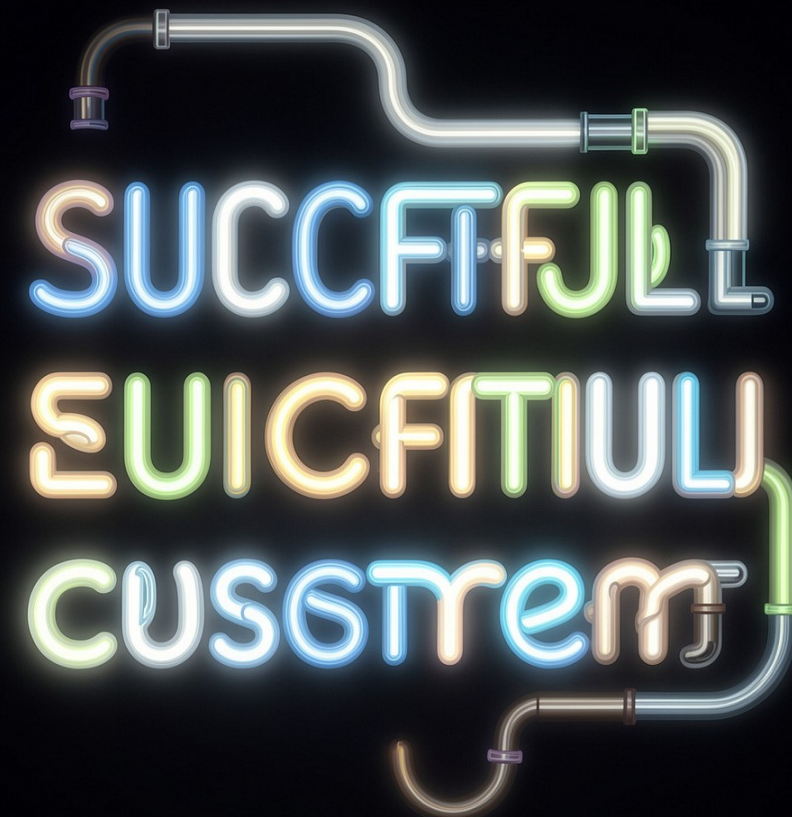
Seamless Management

Transition to automated, paperless record-keeping with instant CSV generation for effortless tracking and reporting.



Modernized Experience

Future-proof your institution with an intuitive, efficient, and technologically advanced attendance solution that sets a new standard.





Results & Performance Metrics

Our system delivers tangible improvements, transforming attendance management into an efficient, reliable, and secure process.

95%+

Accuracy

Consistent facial recognition success rate

<2s

Marking Time

Attendance recorded per student

10+ Min

Time Saved

Recaptured lecture time per session

0

Human Error

Near-zero manual data entry mistakes

The background features a complex network of glowing lines in white, light blue, and orange. These lines connect various circular nodes, some of which are also glowing. The overall effect is a futuristic, interconnected web or circuit pattern.

Thank You